

matlen-et-al-2020-spatial-alignment-facilitates-visual-com

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Paper Synthesis

Not available for this paper.

Abstract

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Humans have a uniquely sophisticated ability to see past superficial features and to understand the relational structure of the world around us. This ability often requires that we compare structures, finding commonalities and differences across visual depictions that are arranged in space, such as maps, graphs, or diagrams. Although such visual comparison of relational structures is ubiquitous in classrooms, textbooks, and news media, surprisingly little is known about how to facilitate this process. Here we suggest a new principle of spatial alignment, whereby visual comparison is substantially more efficient when visuals are placed perpendicular to their structural axes, such that the matching components of the visuals are in direct alignment. In four experiments, this direct alignment led to faster and more accurate comparison than other placements of the same patterns. We discuss the spatial alignment principle in connection to broader work on relational comparison and describe its implications for design and instruction.

Introduction (Paper-Level)

Not available for this paper.

General Discussion

Not available for this paper.

Critique

Not available for this paper.